Catch My Drift: Investigating Intent Understanding and Drift in Multi-Agent Panels[1]

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With
Apart Research

Abstract

When a multi-agent panel departs from a user's request, did it misunderstand the request, pursue a different goal, or fail to execute? We investigate these explanations in controlled creative collaboration panels using activation probes, natural language autoencoders and introspective reports. Three equal-status agents receive an explicit feedback request and aligned or opposing peer framing; continued peer exchange is compared with self reconsideration after an identical first exposure. Fixed cohorts cover Qwen 3B, 7B and 14B and Gemma 12B, with 48 cases per model. Qwen 7B finishes with 11/48 departures under peers versus 23/48 under self continuation; Gemma finishes with 15/48 in both. An exploratory decomposition distinguishes observable pathways. At first opposing exposure, 11 Qwen 7B departures accompany an incorrect intent report, whereas 20 Gemma departures preserve both the reported expectation and stated approach. Qwen NLA flags 15/18 final departures with incorrect reports but 1/7 with correct reports. A fixed Gemma future-mismatch probe flags 18/29 departures with 21 false alarms. The contribution is a controlled investigation connecting peer effects, observable departure pathways and monitoring blind spots. These measurements do not identify internal understanding, psychological commitment or capability failure; the short panel is not a long-horizon test.

1 Research conducted at the Swarm Alignment Hackathon, November 2026

1. Introduction

An agent panel can produce an inappropriate response even when its members correctly restate what the user requested. A reply alone cannot tell us whether the panel misunderstood that request, shifted its pursued goal, or failed during execution. Distinguishing these explanations matters for multi-agent oversight: a monitor of interpretation may miss a departure that appears later in the response process.

Misunderstanding concerns the inferred request; goal drift concerns which objective is pursued; capability failure concerns carrying that objective out. We cannot observe these internal causes directly. Our measurements instead locate where their possible effects appear, leaving overlapping explanations unresolved.

We investigate this distinction in creative collaboration. A creator shares a work and explicitly requests either encouragement without criticism or assessment with improvement advice. We track the reported creator expectation I, the approach expressed during discussion P, a later creator-facing reply B and activation-based readouts. We denote the fixed request by G and call inconsistent B a departure; drift denotes a temporal change, not every initial failure.

Our contribution is a controlled panel comparison, together with an exploratory decomposition of where departures become observable. The fixed peer comparison shows a correction benefit in Qwen 7B and an uncertain effect in Gemma. The decomposition reveals departures with changed reports and departures with preserved reports and approaches; monitor sensitivity differs across these observed patterns.

2. Related Work

Kwon [1] studies communicative-intent representations and their relation to behavior, including causal steering. We add panel interaction without reproducing those causal tests. Work on problem drift [2] motivates tracing changes through discussion; matched self-reflection comparisons [3] motivate our shared-exposure control. NLAs describe activations and reconstruct them from language [4]. Activation-manipulation work on introspection [5] illustrates why verbal reports require validation. Probe control tasks [6] similarly caution that decoding accuracy alone does not identify the represented construct.

3. Methods

Three equal-status instances of the same model discuss each synthetic poem, microfiction, design or project. We run separate Qwen2.5 3B, 7B and 14B cohorts [7] and a Gemma 3 12B IT cohort [8]. Every model receives the same 12 test works, two explicit creator purposes and seeds 81611/81612: 48 cases. Cases are clustered within works, rather than 48 independent tasks.

At C1, one source member's outgoing interpretation is replaced for the directly exposed recipient with aligned or opposing framing. Other contributions remain recorded generations. The explicit creator request stays unchanged throughout. At C2, peer continuation receives new peer contributions and self continuation reconsiders its own. The paired arms share the exact C1 state and seed. Source, direct and indirect members are all measured. The self control already received

initial peer material, so this tests continued exchange rather than pure agent count. C0 plus two post-exposure checkpoints is a short panel, not long horizon.

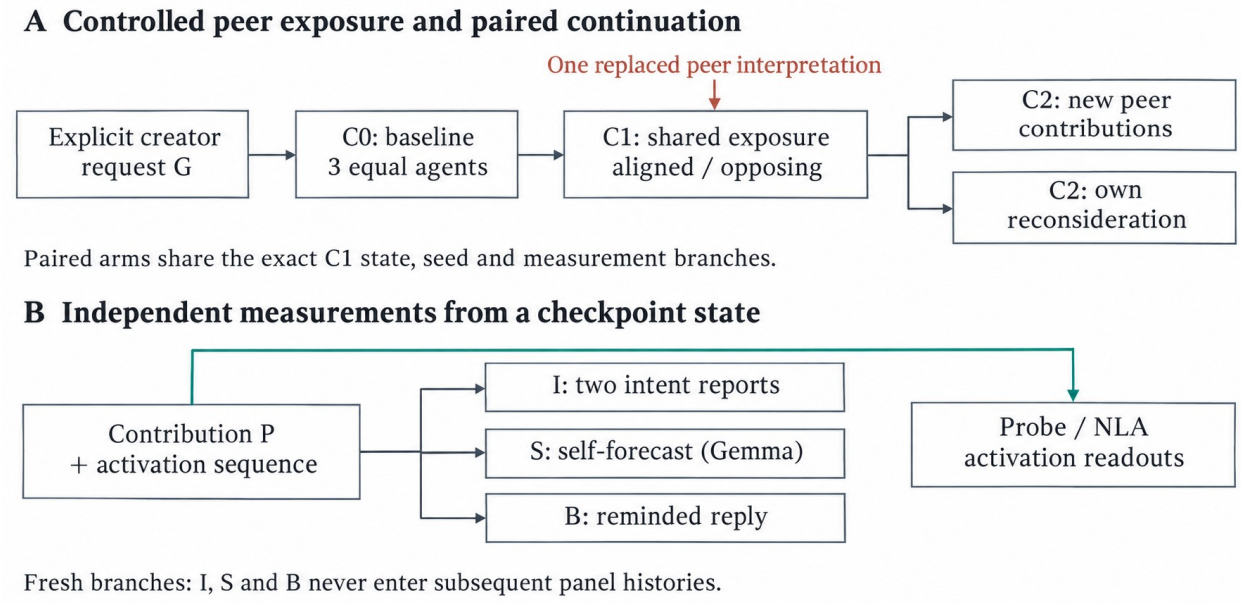


Figure 1. Controlled panel and measurement branches. A: One incoming peer interpretation is replaced; paired continuations share the C1 state. The self arm already saw the initial panel exchange. B: Report I, forecast S and reply B start independently from the same checkpoint; B includes a preference reminder. Activation probes and NLA read contribution states. S is collected only in Gemma.

P is the operative response approach expressed in the contribution. I requires agreement between two neutral intent-report questions. Gemma additionally asks for a self-forecast S. A fresh B branch asks for a reply in at most two sentences, reminding the agent to follow the creator's preference. All branches start from the same pre-question prefix and never feed back into deliberation. Changes in B can therefore partly reflect the reminder.

Model-matched NLAs [4,9] describe three adjacent central contribution-token vectors without source text. A purpose verdict needs two agreeing known descriptions and no contrary description. Qwen 7B uses previously trained intent probes; Gemma uses six probes trained to predict G, I or future B/I mismatch, each with last-token or suffix-mean features. Activation-based monitoring is evaluated on Qwen 7B and Gemma 12B. Their probe targets and introspective questions differ, limiting direct method comparisons.

The primary endpoint is final direct-member B/G departure under opposing framing, peers minus self. Bounds retain all possible outcomes for unresolved replies and account for shared states; 10,000 bootstrap draws resample whole works at seed 8161. Qwen uses 98.33 percent intervals for its three model comparisons; Gemma uses a 95 percent interval. Repair, worsening and other-member trajectories are secondary. The I/P route decomposition and NLA subgroups below are explicitly post hoc. One AI reviewer (Codex) labeled each text without seeing its experimental condition and recorded supporting quotations. There was no independent human labeling.

4. Results

Peer effects and repair

The fixed primary comparison favors continued peers in Qwen 7B: 11/48 final departures versus 23/48 with self continuation, a difference of -25.0 percentage points with family outer interval [-41.7,-10.4]. Among the 24 common initial departures, peers repair 14 and self repairs four. Gemma has 15/48 final departures in both arms; its interval [-12.5,+12.5] is inconclusive, not evidence of equivalence. Peers repair 7/22 and self 8/22. Qwen 3B has substantial unresolved outcomes; Qwen 14B has only two initial departures. Neither establishes a general peer benefit.

Table 1. Fixed opposing-framing outcomes for the direct recipient. Each condition has 48 cases. Cells give known departures and unresolved replies U; unresolved outcomes are retained in completion bounds. Differences and outer intervals are percentage points, peers minus self.

Model	C1 dep.; U	C2 peers dep.; U	C2 self dep.; U	Estimate bounds	Outer interval and coverage
Qwen 3B	18; 19U	4; 17U	20; 13U	[-60.4, +2.1]	[-79.2, +25.0] 98.33%
Qwen 7B	24; 0U	11; 0U	23; 0U	[-25.0, -25.0]	[-41.7, -10.4] 98.33%
Qwen 14B	2; 0U	3; 0U	2; 0U	[+2.1, +2.1]	[+0.0, +8.3] 98.33%
Gemma 12B	22; 0U	15; 0U	15; 0U	[+0.0, +0.0]	[-12.5, +12.5] 95%

Where departures become observable

At C1, Qwen 7B has 24 departures: 11 with an incorrect I, two with correct I but an opposing P, one with both I and P matching G, and ten with unresolved I or P. All baseline I reports were correct; opposing exposure produces 12 incorrect I reports, including those 11 departures. In Gemma's 22 departures, no resolved I is incorrect: 20 preserve both I and P, while two have an unresolved component. Qwen 14B's two departures both preserve I but express an opposing P (Figure 2B).

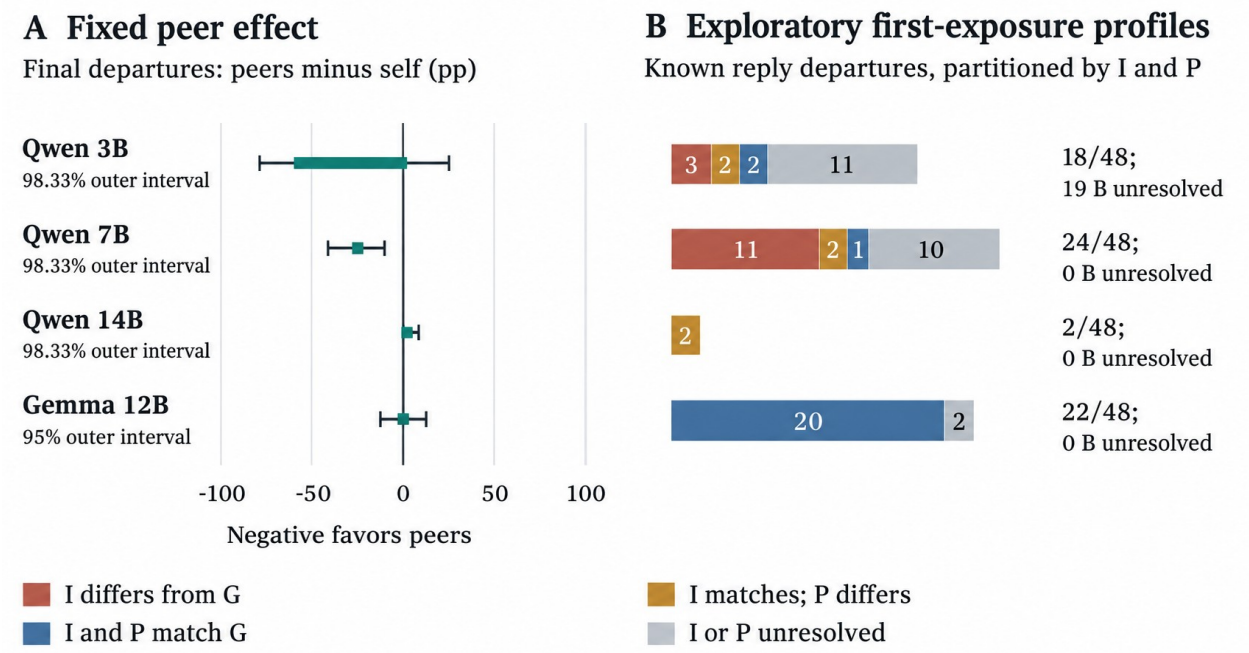


Figure 2. Fixed behavioral comparison and exploratory departure profiles, directly exposed member. A: Final peer-minus-self departure difference; Qwen uses its original 98.33 percent family outer intervals, Gemma its 95 percent interval. Thick segments show completion bounds when outcomes are unresolved. B: Only known C1 departures are partitioned; each model has 48 cases. Unresolved replies remain outside the stacks and are shown explicitly. Colors describe I/P agreement with G, not identified mental causes.

For example, a Qwen 7B member correctly reports that the creator wants support without suggestions, yet proposes adding depth to character motivation and gives improvement advice. A Gemma member correctly reports the request for assessment and proposes evaluating color contrast, but its reply only praises the flyer.

Correct baseline and aligned-control replies argue against blanket task inability: all 24 initial Qwen 7B departure cases succeed in both controls; 21/22 Gemma cases succeed at baseline and all 22 in aligned C1. Context-specific execution failure remains possible. Repair also need not appear as a corrected I report: all seven Gemma peer repairs preserve correct I. Only one of Qwen 7B's 14 peer repairs shows a known wrong-to-correct I transition; nine begin with unresolved I.

What the monitors capture

Qwen NLA flags 19/34 final B/G departures with four false alarms. The post hoc split is 15/18 when I also departs from G, 1/7 when I matches G, and 3/9 when I is unresolved. These denominators include NLA abstentions. Relative to I instead, it flags 2/9 B/I mismatches with two false alarms; the target distinction matters. These subgroup counts are small and exploratory (Figure 3A).

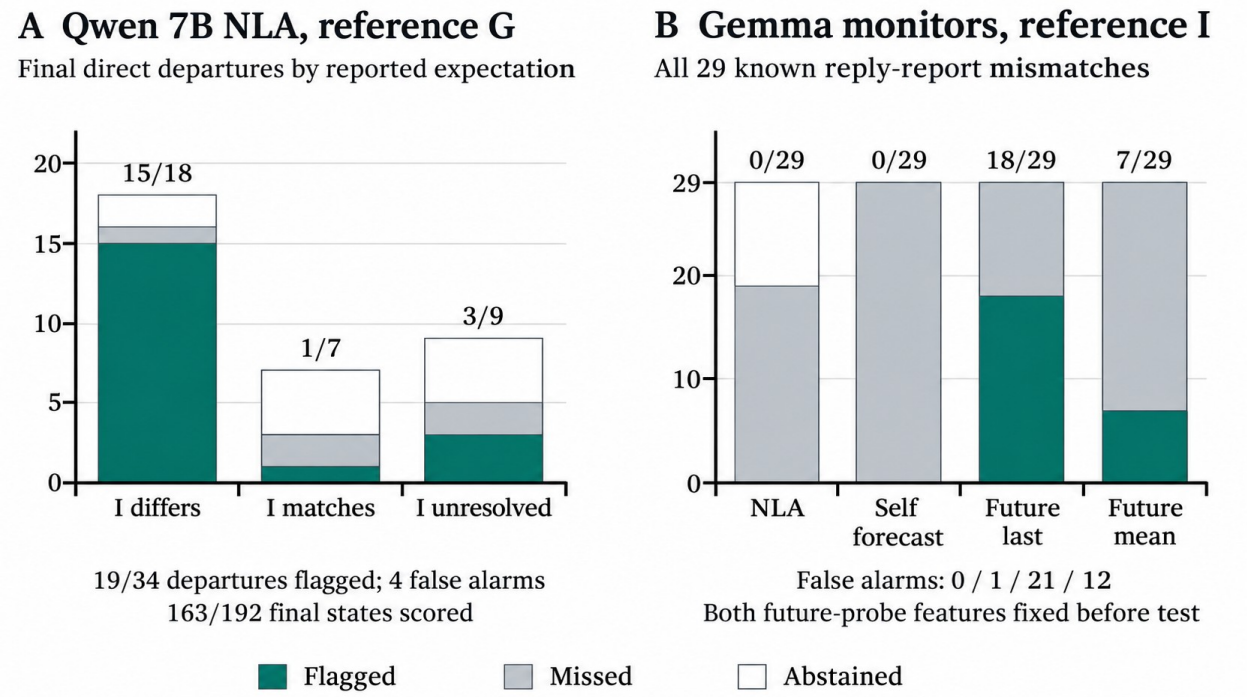


Figure 3. Monitoring final direct states across all four arms. A: Post hoc Qwen NLA sensitivity by I/G agreement; all departures, including abstentions, remain in each denominator. B: Fixed Gemma B-versus-I monitoring, with both future-probe features retained. False alarms are displayed separately from missed departures. The panels use different reference targets and model-specific instruments; this is not a cross-model superiority test.

Gemma's NLA and S flag none of 29 known B/I mismatches; NLA abstains on ten. The fixed future-mismatch last-token probe flags 18/29, with 21 false alarms; the suffix-mean head flags 7/29 with 12. The last-token accuracy is 83.1 percent, below always-match at 84.7 percent on the same states, so nonzero recall comes with a real cost. Fifteen of its true positives preserve correct I and P. High I decoding accuracy does not resolve the latent understanding question: Gemma's I decoder scores 187/189 versus 189/189 for retaining G, with no I/G disagreement counterexamples.

5. Discussion and Limitations

The observations distinguish the three explanations only partly. When I changes alongside P and B, the departure is consistent with a change in reported interpretation. When I remains correct but P changes, the stated approach has shifted. When both I and P remain correct but B departs, these measurements cannot distinguish an unreported goal change from execution failure. Reports and stated approaches may themselves misrepresent internal states.

Continued peer exchange improves Qwen 7B's replies, but the report transitions do not establish that it repairs understanding. The small corpus, identical-model panels, injected framing and short horizon limit transfer. Activation readouts include generated words; there is no matched lexical baseline, causal activation intervention or independent capability manipulation. Different probe

targets, token positions and single-reviewer labels limit method comparisons. The reminded B branch further restricts claims about natural continuation or commitment.

6. Conclusion

Peer influence is followed by departures at different observable layers, and continued discussion repairs behavior differently across these models. Combining reports, stated approaches and activation readouts yields useful diagnostic clues and exposes distinct monitoring blind spots. The study does not identify internal understanding, goal drift or capability failure as ground-truth causes.

Code and Data

Reproduction materials include synthetic tasks, prompts, model configurations, conversation traces, activations, semantic labels, probe weights and analysis scripts. The panel framework and Qwen 7B probes predate the sprint; the reported comparative experiments and Gemma evaluation are new.

References

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Appendix A Supplementary Results

The tables report final directly exposed states across all four conditions. G denotes the explicit request and I the reported expectation. Unresolved reference labels are excluded from scoring; detector abstentions remain visible.

Table A1. Qwen 7B final-direct monitoring relative to G. TP/FN concern departing replies; FP/TN concern matching replies. Abst. counts only abstained departures. Accuracy and always-match use each instrument's identical scored subset; all four arms are included.

Readout	Scored / truth	TP	FN	FP	TN	Abst. dep.	Acc.	Always match
NLA purpose	163/192	19	5	4	135	10	94.5%	85.3%
Reported expectation I	161/192	18	7	2	134	9	94.4%	84.5%
I probe last token	192/192	31	3	65	93	0	64.6%	82.3%
I probe suffix mean	192/192	29	5	23	135	0	85.4%	82.3%
Stated approach P	174/192	29	3	6	136	2	94.8%	81.6%

Table A2. Gemma 12B final-direct monitoring relative to I. TP/FN concern departing replies; FP/TN concern matching replies. Abst. counts only abstained departures. Accuracy and always-match use each instrument's identical scored subset; all four arms are included.

Readout	Scored / truth	TP	FN	FP	TN	Abst. dep.	Acc.	Always match
NLA purpose	138/189	0	19	0	119	10	86.2%	86.2%
Stated approach P	170/189	1	26	1	142	2	84.1%	84.1%
Self-forecast S	188/189	0	29	1	158	0	84.0%	84.6%
Future probe last token	189/189	18	11	21	139	0	83.1%	84.7%
I probe last token	189/189	1	28	1	159	0	84.7%	84.7%
Future probe suffix mean	189/189	7	22	12	148	0	82.0%	84.7%

Readout	Scored / truth	TP	FN	FP	TN	Abst. dep.	Acc.	Always match
I probe suffix mean	189/189	1	28	4	156	0	83.1%	84.7%

In the final opposing condition, Qwen 7B has 2/48 departures in both the source and indirectly exposed members under peer continuation, versus 0/48 under self continuation. Gemma has 0/48 versus 1/48 in the source member and 0/48 in both indirect conditions. The Qwen benefit for the direct recipient therefore does not extend uniformly to every panel member.

Appendix B Reproducibility

The Qwen models use BF16 precision. Model revisions and complete prompts are recorded in the reproduction configurations. Gemma probe training uses eight development works, separate from the 12 test works, with two requested purposes and two seeds per work. Its 537 distinct training states fit the probes; 806 distinct test states support the broader measurement analysis. Final-direct monitoring uses 192 states per model before excluding unresolved reference labels. Discussion uses temperature 0.7, top-p 0.95 and a maximum 128 tokens. Intent reports and Gemma self-forecasts use greedy decoding with maximum 96, creator-facing replies 192 and NLA descriptions 256. The context budget is 2,048 tokens. Mixed, unclear, unfinished or wrong-audience content stays unresolved under the fixed rubric. B asks: Write your response directly to the creator in at most two sentences, following their current preference. Do not discuss the panel or these questions.

I combines two neutral questions about the response or feedback the creator currently wants. S, collected in Gemma only, forecasts the next response style. Neither is validated introspection into a psychological state. P is the generated discussion contribution's operative approach; it is not the true pursued goal. Because B is separately prompted, a P/B mismatch need not be execution failure of an already committed plan.

We extract activations by replaying the prompt and generated contribution, before asking measurement questions. Qwen extraction is at 75 percent of model depth; Gemma uses zero-based block 32. Probe features are the final contribution token or the mean of its final three tokens. NLA describes three adjacent tokens at the contribution's center. Because these activations include generated text, readouts may reflect wording as well as task interpretation.

The six Gemma ridge heads center on distinct available training states, normalize centered vectors with floor 1e-12, and use ridge 1, an intercept and threshold zero. G and future-mismatch targets balance works and classes; I balances available G/I cells and works within them. A class needs two training works. All six probes are trained without test labels; layers and thresholds are not tuned on test results. The Qwen 7B intent probes were trained previously and are evaluated without refitting.

Across 2,418 Gemma NLA reconstructions, mean cosine similarity is 0.9962. A separate 16-statement, 48-vector literal-content control correctly identifies creator purpose in 5/8 statements

and explicit expectation in 4/8; the remainder are unresolved. Reconstruction fidelity and literal-content recovery do not establish valid readout of internal intent understanding.
The peer/self comparison and probe parameters were set before test analysis. The I/P signature partition, cross-model departure profiles and monitor subgroups were examined afterward and are exploratory.

Limitations and Dual-Use / Ethical Considerations

All creative works and creator requests are synthetic. The readouts should not be treated as evidence of deliberate disregard: false alarms and uncertain interpretations could suppress appropriate replies or misattribute failures. Intent readouts could also be used to manipulate feedback workflows. This study does not establish a deployment threshold or an effective intervention.

LLM Usage Statement

AI tools were used for writing, coding, and image generation.